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Auditory–Articulatory Neural Alignment between Listener and Speaker during Verbal Communication

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Abstract

Whether auditory processing of speech relies on reference to the articulatory motor information of speaker remains elusive. Here, we addressed this issue under a two-brain framework. Functional magnetic resonance imaging was applied to record the brain activities of speakers when telling real-life stories and later of listeners when listening to the audio recordings of these stories. Based on between-brain seed-to-voxel correlation analyses, we revealed that neural dynamics in listeners' auditory temporal cortex are temporally coupled with the dynamics in the speaker's larynx/phonation area. Moreover, the coupling response in listener's left auditory temporal cortex follows the hierarchical organization for speech processing, with response lags in A1+, STG/STS, and MTG increasing linearly. Further, listeners showing greater coupling responses understand the speech better. When comprehension fails, such interbrain auditory-articulation coupling vanishes substantially. These findings suggest that a listener's auditory system and a speaker's articulatory system are inherently aligned during naturalistic verbal interaction, and such alignment is associated with high-level information transfer from the speaker to the listener. Our study provides reliable evidence supporting that references to the articulatory motor information of speaker facilitate speech comprehension under a naturalistic scene.

Key words: auditory–articulatory interaction, interbrain neural coupling, speech comprehension

Introduction

Verbal communication by nature is the transfer of information across brains. Though the acoustic property of speech produced by a speaker is complex and highly variable, a listener can typically decipher the linguistic information quickly and effortlessly. Neural mechanisms underlying this process remain

incompletely understood. Cognitive models propose that auditory processing of speech is facilitated by references to the intended articulatory gesture of the speaker (Liberman and Mattingly 1985; Galantucci et al. 2006). Indeed, it has been reported that motor cortex is involved in speech perception. Evidence includes 1) the motor cortex responds to clear and degraded speech (Watkins et al. 2003; Wilson et al. 2004;

Pulvermüller et al. 2006; Du et al. 2016); 2) transcranial magnetic stimulation of motor cortex interferes with phonological discrimination tasks (Meister et al. 2007; Möttönen and Watkins 2009). However, there is argument that such a co-involvement of motor cortex might be epiphenomenal reflecting spreading activation between associated networks or might be related to task-specific motor control (Hickok 2009, 2014; Stassenko et al. 2015; Arsenault and Buchsbaum 2016). So far, whether articulatory motor information is utilized to facilitate speech comprehension in daily life situation remains elusive.

While existing studies are mostly focused on the neural response within the boundary of individual brains, recent perspectives from psycholinguistics and cognitive neuroscience propose that the neural mechanism underlying speech processing should be more completely understood within a two-brain framework (Pickering and Garrod 2013; Hasson and Frith 2016). Indeed, the proposed auditory-motor linkage does not emerge in isolation within a listener's brain but builds upon the intensive interaction of the listener with speakers during human language evolution and speech acquisition (Liberman and Mattingly 1985). Thus, to fully understand the linkage between auditory system and motor system and its functional role in speech comprehension, it is necessary to examine it under naturalistic verbal interactions in the first place.

Two lines of previous findings indicate that the auditory cortex of a listener may be inherently aligned with a speaker's articulatory cortex during verbal communication. Firstly, the output sound is closely associated with the activity in the speaker's articulatory cortex (sound-articulatory alignment; Magrassi et al. 2015; Perez et al. 2017). Secondly, the temporal features of the speech sound are tracked by the neural dynamics of listener's auditory cortex (sound-auditory alignment; Ding and Simon 2012; Vander Ghinst et al. 2016). Although it is speculated that there should be a close link between listener's auditory system and speaker's motor system (i.e., auditory-articulatory alignment), this hypothesis has never been directly tested, and its potential functional role in verbal communication remains unknown. Brain-to-brain alignment can be indexed by the coupled responses across brains over time. Previous studies have documented that responses between the homologous cortical areas of listener and speaker are coupled during successful communication (Stephens et al. 2010; Jiang et al. 2012; Kuhlén et al. 2012; Silbert et al. 2014; Liu et al. 2017). However, the relation between the auditory system in listener and articulatory system in speaker, two systems with seemingly quite divergent functions, has never been explicitly investigated.

To test the above hypothesis, we scanned speakers telling real-life stories and latter listeners hearing audio recordings of the stories using functional magnetic resonance imaging (fMRI). Then neural couplings between listener's auditory cortex and speaker's articulatory cortex were calculated. We explored a set of articulatory regions in the speaker's brain presumed to play distinct functional roles in speech production, with a primary interest in the larynx/phonation area (LPA), which is thought to be specific for phonation (Brown et al. 2008). For listeners, we were interested in the auditory system in broad sense, including early auditory cortex (A1), superior temporal gyrus/sulcus (STG/STG) and middle temporal gyrus (MTG). If the coupling response to a speaker's articulatory cortex plays a functional role in speech comprehension, the A1, STG, and MTG should present different temporal profiles in their coupling responses, given that these regions are thought to be engaged in different stages of speech processing (Oleser et al. 2007; Peelle et al.

2010; Friederici 2012). Also, the coupling response of auditory cortex should be enhanced when the speech is understandable. Further, if the neural coupling between the listener's auditory cortex and speaker's articulatory cortex serves as a mechanism underlying information transfer from the speaker to listener, the strength of coupling between a pair of conversers should predict the success of communication.

Materials and Methods

Participants

Two female speakers (one Chinese and one Mongolian) and 43 native-Chinese listeners (23 females, ages 20–75 years) participated in this study. The Chinese spoken by these participants is a dialect used in Chongqing, which is very close to Mandarin. Speakers were chosen by a loose “audition” to ensure they could talk fluently. No listener could understand Mongolian. All participants were right-handed based on Edinburgh Handedness Inventory (Oldfield 1971) and reported no neurologic or mental disorders. For participants older than 53 years, a mini-mental state examination was administrated. All of them scored >26 on this examination. A digit memory test was administrated to assess participants' memory capacity (Wechsler 1987). Digit span was measured by averaging the number of digits correctly repeated forwards and backwards. Pure tone audiometry was used to measure participants' hearing sensitivity. We used a threshold of < 30 dB HL in better ear across 0.5, 1, 2, and 4 kHz to filter the participants (Peelle and Wingfield 2016), which are the frequencies covering the most range of human speech (Turner and Cummings 1999). All participants met the threshold at 0.5, 1, and 2 kHz, and all but two met at 4 kHz. Written informed consent was obtained from all participants under a protocol approved by the Reviewer Board of Southwest University.

Experimental Design

Both the Chinese and Mongolian speakers told real-life stories (10 min long) while undergoing fMRI scanning. Their speech was recorded in real time using a noise-canceling microphone (FOMRI-III; Optoacoustics Ltd, Or-Yehuda, Israel), which was positioned above the mouth of the speaker. To make the language production more natural, the speakers were told that somebody was listening to them outside, and they were instructed to tell their stories in a way that could appeal the audience and keep their heads as still as possible. The audio recordings were processed offline using Adobe Audition 3.0 (Adobe Systems Inc., USA) to further remove fMRI scanner noise. Since the Chinese speaker had a sudden cough at the last minute of the recording, we played back only the first 7-min recordings of both Chinese and Mongolian stories to each of the 43 listeners during fMRI scanning. The auditory stimuli were presented through a MRI-compatible headphone with noise-cancellation (OptoACTIVE, Optoacoustics Ltd, Or-Yehuda, Israel) at a volume comfortably audible for all listeners. During scanning, a same set of visual stimuli, consisting of a fixation cross lasting for 20 s and then an icon of a horn lasting until the end of the session, were presented to speakers and listeners. The horn was designed to prompt the participant to start speaking or listening immediately. The presentation of auditory and visual stimuli was controlled by Psychtoolbox-3 running on Matlab R2012b (MathWorks, Natick, MA, USA). Half of the participants listened to one Chinese story and the other

half listened to another Chinese story. The reason of using two different stories here was to avoid a potential repetition effect since the participants also ran another experiment that was not reported in this study. As participants listening to the two different stories did not differ in their comprehension score (two sample *t*-test, $P_{(20, 21)} = 0.65$), their data were collapsed into one group for further analyses. To make sure that listeners attended to the Chinese story, they were informed beforehand that an interview about the content of the story would be given after the scanning. In another run, the Mongolian story was played back to all listeners, and participants were instructed to listen to the story attentively even though they could not comprehend it.

Behavioral Assessment

Immediately after the scanning session, participants were asked to retell the story with as much detail as possible. After the free recall, the experimenter asked the listeners several questions regarding the part of the story not present in their recall. Two independent raters scored each listener based on the recording of their free recall and their answers to the experimenter's questions. To make the scoring as objective as possible, we made a list of questions concerning the important contents of the story. The correct answer to each question contained several key points including character, place, time and thought, etc. The raters scored the participants on each question based on the information they provided on those key points. The final score was a ratio of the summed score over all questions to full score. The assessments made by the two raters were in high agreement (Spearman's correlation coefficient, 0.89). The mean score of the two raters was used as a quantitative measure for listener's comprehension level.

MRI Acquisition and Preprocessing

Images were acquired with a 3T Siemens Trio Scanner at the MRI Center of the Southwest University in China. A gradient echo planar imaging sequence was applied for functional images acquisition. Parameters were as follows: repetition time, 2000 ms; echo time, 30 ms; flip angle, 90°; field of view, 220 mm; matrix size, 64 × 64; slice number, 32 interleaved; and voxel size, 3.44 × 3.44 × 3.99 mm³. A magnetization prepared rapid gradient echo sequence was applied for anatomical images acquisition, with the following parameters: repetition time, 2530 ms; echo time, 3.39 ms; flip angle, 7°; FOV, 256 mm; scan order, interleaved; matrix size, 256 × 256; and voxel size, 1.0 × 1.0 × 1.33 mm³.

Two hundred and ten volumes were acquired for each run. The first 10 volumes corresponding to the fixation period were discarded. The DPABI toolkit (Yan et al. 2016) based on SPM12 (www.fil.ion.ucl.ac.uk/spm/) was utilized for image preprocessing. The raw functional images were slice-time corrected, spatially realigned and co-registered to their corresponding anatomical images, normalized to Montreal Neurological Institute (MNI) space, resampled into 3 × 3 × 3 mm³ voxel size and smoothed (full width at half maximum (FWHM), 7 mm). The resultant images were further detrended, nuisance variable regressed, and high-pass filtered (1/128 Hz). The nuisance regression included removal of five principal components of white matter and cerebrospinal fluid within individual subject's T1 segmentation mask (Behzadi et al. 2007), as well as Friston's-24 motion parameters (including each

of the six motion parameters of the current and preceding volume, plus each of these values squared; Friston et al. 1996).

Data Analyses

The dataset of one participant with extremely low performance (comprehension score <6%) was discarded. In addition, the datasets of another two participants with excessive head movement (>3 mm or 3°) were discarded. The final analysis was performed on 40 participants. Datasets of the two participants who had moderate hearing loss at 4 kHz were included in the final analysis considering their normal task performance (comprehension score for both was 61.46%, corresponding to a *z* score of −0.83).

ROI and Temporal-Cortex Mask Definition

We conducted between-brain seed-to-voxel correlation analyses to assess the neural coupling of auditory temporal cortex in the listener to the articulatory seeds in the speaker. Four seed regions of interest (ROIs) for the articulatory system were chosen *a priori*, based on activation peaks reported previously (Brown et al. 2008). In that study, both glottal stops and phonation tasks were examined. Since much of speech signal is voiced, we chose the reported peaks for the phonation task. The chosen peaks included 1) the left and right ventromedial LPA (MNI coordinates: 44, −10, 36; −40, −12, 32), which are thought to control the movement of larynx; 2) the left ventral precentral gyrus (MNI coordinates: −54, 2, 26), thought to control the movement of the tongue; and 3) the left frontal operculum (MNI coordinates: −60, 6, 12), a high-level articulatory region likely related to articulation planning. Spheres with a radius of 6 mm surrounding these peaks were constructed. Talairach coordinates reported in the original study were converted to MNI coordinates following the same procedure as used in that study for MNI-to-Talairach conversion. Note that the LPA coordinates we chose matched quite well with those reported by a previous meta-analytic study on oral reading (Turkeltaub et al. 2002; with a distance of 2 mm for right LPA and 8.2 mm for left LPA) and with those reported by a research that calculated the mean coordinates of LPA across several studies (with a distance of 2.4 mm for right LPA and 4.1 mm for left LPA; Grabski et al. 2012). The LPA is our primary interest, since it is thought to control the organ (the larynx) for vocalization and pitch and volume modulation (D'Ausilio et al. 2011). For comparison purpose, we also examined the "tongue" area and the frontal operculum, which are thought to be involved in articulation but are not phonation-specific (Brown et al. 2009).

Analyses for listener's neural coupling to the speaker's articulatory cortex were confined to the auditory temporal cortex that is supposed to be responsible for auditory language processing throughout this study (Price 2012). A mask for the auditory temporal cortex was defined anatomically using Marsbar (<http://marsbar.sourceforge.net/>) based on the Automatic Anatomical Labeling (AAL) atlas, comprising bilateral Heschl's gyrus, superior and MTG, and temporal pole (Fig. 1A). The inferior temporal gyrus was not included since this area is more closely involved in visual rather than auditory processing (Gross 1994).

Estimation of Neural Coupling

To measure neural coupling, time courses of all voxels within the seed ROI from the speaker were averaged. Then Pearson's correlation between the ROI time course and the time course of each voxel within the mask of auditory temporal cortex in the

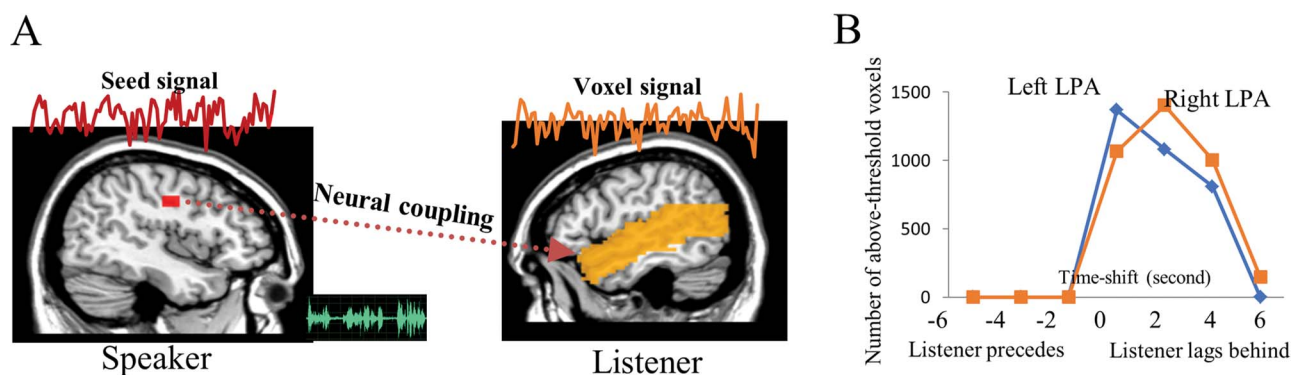


Figure 1. Experimental and analytical protocol. (A) Brain activities of a speaker when telling real-life stories and later of listeners when listening to the audio recordings of these stories were measured using fMRI. The listener–speaker neural coupling was assessed via a between-brain seed-to-voxels (within the mask of auditory temporal cortex) correlation analysis. (B) To capture asynchronous interaction, listeners’ time series were shifted to precede or lag behind those of speaker. Extent of listener–speaker neural coupling at different time shifts was quantified by the number of voxels in listeners’ auditory temporal cortex showing significant coupling (at group level) to speaker’s LPA. There was no significant listener-preceded neural coupling. At lag 6 s, the extent of listener–speaker neural coupling declined dramatically.

listener was computed. The resulting correlation coefficient was normalized using a Fisher’s r -to- z transformation. As listeners may need time to extract the information embedded in the input, their neural processes could lag behind those of the speaker. The listeners may also anticipate what comes next; therefore, their neural processes could precede those of the speaker. For these reasons, we repeated the neural coupling analyses by shifting listener’s time course relative to that of the speaker from -6 s (listener preceded) to 6 s (listener lagged behind) in steps of 2 s. At shift zero, listener’s brain activity was time-locked to the speaker’s vocalization.

Statistical Analyses

For group statistical inference, the individual Fisher’s Z -maps were fed into a random-effect one-sample t -test implemented by DPABI. To avoid arbitrarily setting cluster size thresholds and smoothing levels, Threshold-Free Cluster Enhancement (TFCE) was applied for statistical inference (Smith and Nichols 2009; Chen et al. 2018). A significance threshold was set to PFWE (family wise error) < 0.05 corrected for multiple comparisons across the search volume (i.e., the bilateral auditory temporal cortex). To test the robustness of results, we also conducted a nonparametric permutation test. In this approach, surrogate data was generated by randomly shuffling the time course of each voxel within listener’s auditory temporal cortex. Next, Pearson’s correlation between the surrogate time courses from the listener and the measured time course from the speaker was computed. A null distribution of correlation values was generated by repeating this procedure for 5000 times. Statistical significance was determined by comparing the observed correlations to those correlations obtained from the randomly permuted datasets.

Correlation with Behavior

A previous study has shown that listeners with greater (mirror) neural couplings to a speaker understand speech better (Stephens et al. 2010). Here, we tested whether a positive correlation between comprehension level and auditory-to-articulatory coupling could be found. For each listener, neural couplings (the Fisher’s Z value) to speaker’s LPA at lag 0 s, lag 2 s, and lag 4 s were averaged to represent an integrated measurement for coupling strength. Then a partial Spearman correlation analysis

was applied to assess the relation between listeners’ neural coupling strength of each voxel and their comprehension scores, with the effects of memory capacity (measured by the digit span), age, and hearing sensitivity controlled. These variables are suggested to have a potential influence on a listener’s speech perception and comprehension (Humes and Dubno, 2010) but are not the primary interest of this study. The hearing sensitivity was the average auditory threshold of left and right ears over 0.5 , 1 , 2 , and 4 kHz. Statistical results were corrected for multiple comparisons across the search volume using TFCE, controlling family-wise error rate (FWER) at 0.05 .

Levels of Information Associated with the Auditory–Articulatory Neural Coupling

The auditory–articulatory neural coupling we may observe might be simply caused by shared sensory-level inputs between the speaker and listener or instead be associated with linguistic information transfer from the speaker to the listener. To test these possibilities, we had the participants listen to a story told by a Mongolian speaker. Under this condition, listeners and the speaker shared sensory inputs, but there was no linguistic information transfer between them. We calculated the neural coupling between listeners and the Mongolian speaker using the same procedure described above.

Controlling for the Potential Effects of Neural Coupling between Homologous Areas and Intra-brain Functional Connectivity

Previous studies have found that, when producing speech, speakers’ auditory cortex showed similar responses to those in the listener, as they can hear the same voice (Stephens et al. 2010; Silbert et al. 2014). Meanwhile, due to auditory feedback, brain activities in speakers’ motor system and their auditory system are functionally connected (Christoffels et al. 2010; Behroozmand et al. 2015). In this way, neural dynamics in speakers’ auditory cortex could be correlated to both their own LPA and listener’s auditory cortex. The neural coupling between speaker’s LPA and listener’s auditory cortex, if we find, might be an indirect reflection of the two links (Supplementary Fig. S1 for schematic). To test this possibility, we applied a partial correlation analysis to assess the coupling between speaker’s LPA and each voxel in listener’s auditory temporal cortex, with the neural dynamics in the corresponding voxel of

Speaker's Brain

Listener's Brain

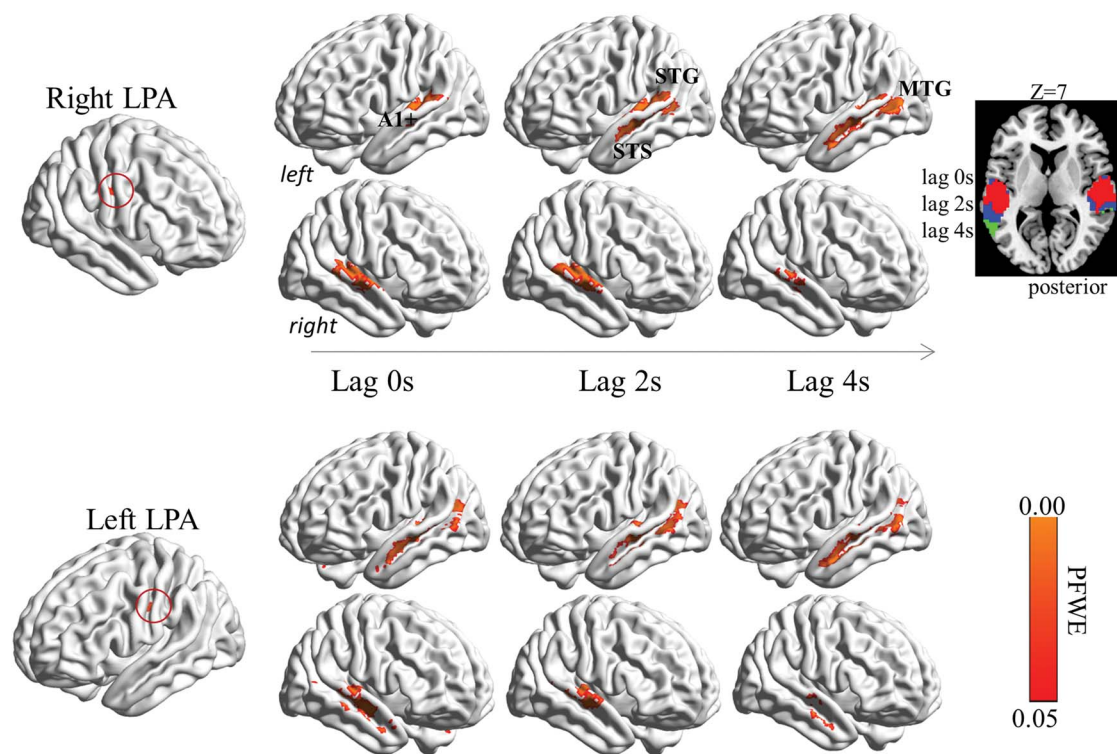


Figure 2. Significant neural coupling between listeners' auditory temporal cortex and speaker's LPA. Notably, from low-order to high-order auditory areas in the left hemisphere, the lag of coupling response increases linearly: 0 s for A1+, 2 s for STG/STS, and 4 s for MTG. The top-right plot shows a typical slice demonstrating this pattern. In this plot, the binarized maps for lag 0 s (red), lag 2 s (blue), and lag 4 s (green) were overlaid sequentially on the Ch2better template using MRIcron software, with 100% transparency on other overlays. A1+: early auditory region; STS: superior temporal sulcus. Threshold: FWER corrected $P < 0.05$ using TFCE. The 3D surface visualizations of the results are implemented using the BrainNet Viewer (www.nitrc.org/projects/bnv).

speakers' auditory cortex being the covariate. The time points of the covariate time series were aligned with those of the speaker's LPA.

On the other hand, when comprehending speech, listeners' motor cortex may have similar responses to those of the speaker due to the hypothesized mirroring mechanism (Pulvermüller and Fadiga 2010). Meanwhile, listeners' motor cortex interacts with their own auditory cortex (Hickok et al. 2003; Möttönen et al. 2012). In this way, neural dynamics in listeners' LPA might be correlated to both speaker's LPA and their own auditory cortex, resulting in a potentially misleading relationship between listener's auditory cortex and speaker's LPA (Supplementary Fig. S1 for schematic). To test this possibility, we also conducted a partial correlation analysis to assess the auditory-articulatory neural coupling between listener and speaker, with neural dynamics in listeners' LPA (at ROI level) being the covariate. Here, the time points of the covariate time series were aligned with those of the listener's auditory cortex.

Results

Behavioral Performance

The average comprehension score of participants was 75.48% ($N = 40$; ranges 37.5–98.4%; $SD = 16.81\%$). Self-report revealed that participants attributed unrecalled content dominantly to a memory problem, rather than to comprehension difficulty. The

post-scan interview for the Mongolian story comprehension task showed that participants had attempted to comprehend the story but failed.

Listener's Auditory Area Was Coupled Selectively to Speaker's LPA

Neural couplings between the articulatory seeds in speaker and each voxel within the auditory temporal cortex in listeners, with delays in listeners' brain varying from -6 to 6 s, were assessed. When the participants listened to the Chinese story, neural dynamics in their auditory temporal cortex were significantly coupled with the neural dynamics in speaker's bilateral LPA, synchronously or with lags. No significant neural coupling to either left or right LPA was found when listeners' time courses were shifted to precede those of speaker. At lag 6 s, the extent of listener-speaker neural coupling declined dramatically (Fig. 1). Therefore, subsequent part only reports the results for lag 0, 2 and 4 s.

Spatial maps for these significant speaker-listener neural coupling are shown in Fig. 2. Remarkably, the coupling response to speaker's right LPA follows an orderly temporo-spatial progression in listeners' left hemisphere: starting in the A1+ (listener synchronized with speaker), then spreading to STG/STS (listener lagged by 2 s), and eventually spreading to MTG (listener lagged by 4 s). This temporo-spatial progression was

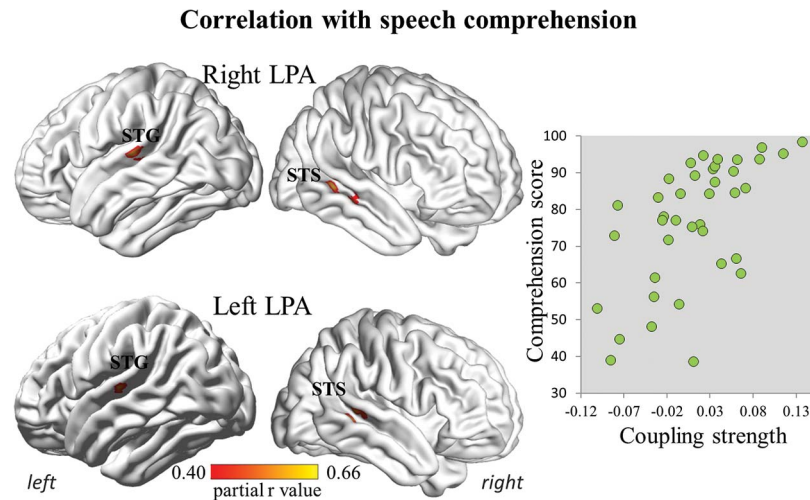


Figure 3. Listener with greater neural coupling understood the speech better. The scatter plot shows distributions of listener's comprehension score against the coupling strength of a representative voxel (MNI: $-50, -34, 1$). Threshold: FWER corrected $P < 0.05$ using TFCE.

accompanied by increasingly left-hemisphere dominance. Results obtained from the analysis on the datasets that excluded the two participants with moderate hearing loss at 4 kHz was similar to the results that included them (Supplementary Fig. S6). Results obtained from the permutation test also showed a similar picture (Supplementary Fig. S7).

No significant neural coupling between listener's auditory temporal cortex and speaker's "tongue" area or left frontal operculum was found.

Stronger Coupling Response Was Associated with Better Comprehension

To explore the functional role of coupling response in listener's auditory cortex to speaker's articulatory cortex, we estimated the correlation between listeners' mean coupling strength across lag 0 s, lag 2 s, and lag 4 s and their comprehension scores. The coupling responses in listener's STG/STS to speaker's right LPA were positively correlated with comprehension score. A similar pattern was found for the coupling responses to speaker's left LPA (Fig. 3). Information for significant clusters is provided in supplementary material (Supplementary Table S1).

Coupling Response Largely Vanished When Comprehension Failed

When listening to the Mongolian story (failed comprehension), listeners showed only significant neural coupling to speaker's left and right LPA in the early auditory cortex (the Heschl's gyrus and nearby regions; Fig. 4).

The comparison between listener's couplings to the Chinese speaker (successful comprehension) and to the Mongolian speaker (failed comprehension) revealed that the synchronized coupling response in the early auditory cortex did not differ between the two conditions. Stronger coupling to the Chinese speaker than to the Mongolian speaker was mainly found in listeners' high-order auditory areas (STG/STS and MTG; Fig. 5). Together, these results suggest that the coupling responses in listener's early auditory area to speaker's LPA arises from shared sensory-level acoustic input, while the coupling of listeners'

Neural coupling with unintelligible speaker

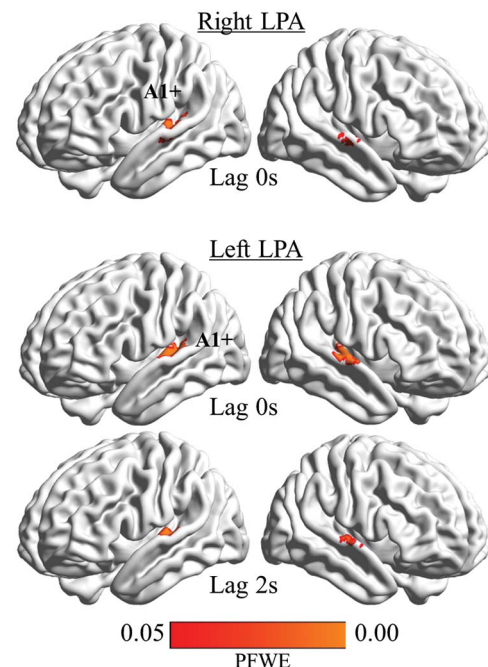


Figure 4. Listeners' neural coupling to speaker's LPA when comprehension failed. No significant listener-speaker neural coupling at lag 4 was found. Threshold: FWER corrected $P < 0.05$ using TFCE.

high-order auditory areas with speaker's LPA is likely associated with high-level linguistic information transfer from the speaker to the listener.

Auditory-to-Articulatory Coupling Was Not an Epiphenomenon of Intra-brain Functional Connectivity and Inter-brain Mirroring Couplings

We observed significant auditory-motor functional connectivities within the brains of both listeners and speaker

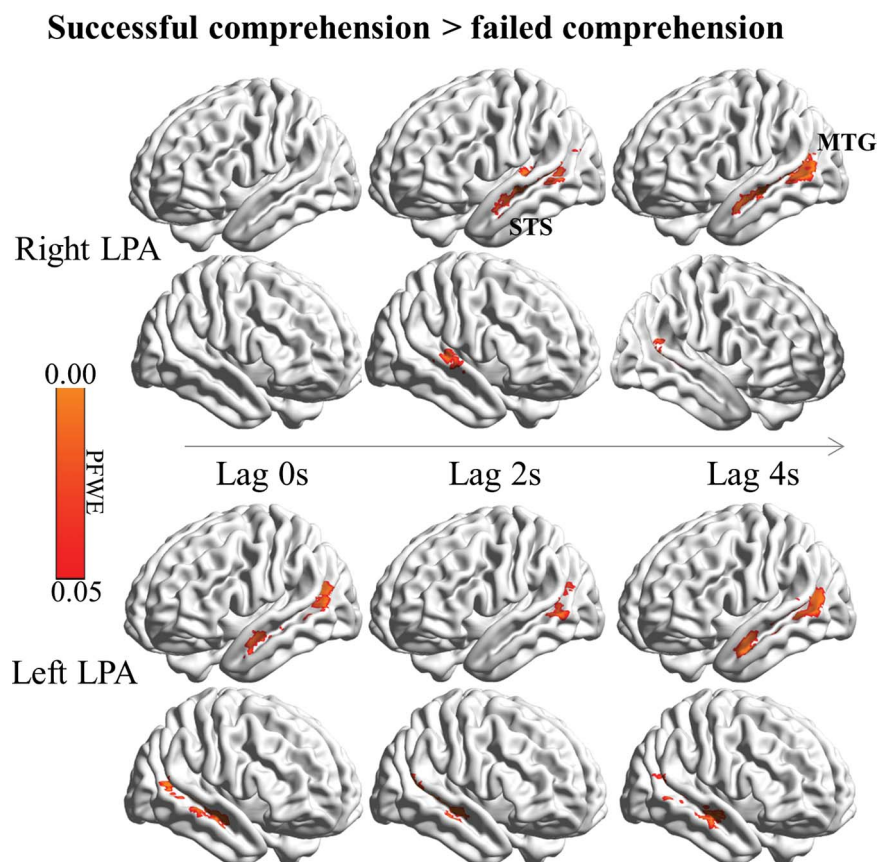


Figure 5. Comparison between listener's neural coupling to the Chinese speaker (successful comprehension) with that to the Mongolian speaker (failed comprehension). Differences between the two conditions were mainly shown in high-order auditory areas (STS, MTG). Threshold: FWER corrected $P < 0.05$ using TFCE.

(Supplementary Fig. S2). We also found significant auditory–auditory neural coupling between listener and speaker (Supplementary Fig. S2). There was no significant LPA–LPA neural coupling between listener and speaker (Supplementary Fig. S3). After controlling for these effects, there was still significant auditory–articulatory neural coupling between listener and speaker, with a similar temporal-spatial profile as revealed by the original analysis (Supplementary Figs S4 and S5). These findings suggest that the coupling of listeners' auditory cortex to speaker's articulatory cortex was not merely an epiphenomenon for the co-presence of intra-brain auditory–motor functional connectivity and inter-brain mirroring responses.

Discussion

The present study revealed that, during the comprehension of naturalistic speech, neural dynamics in listener's auditory cortex were consistently coupled with neural dynamics in the speaker's articulatory cortex. This between-brain auditory–articulatory coupling was independent from intra-brain auditory–articulatory functional connectivities and inter-brain mirror couplings. Interestingly, the coupling response in listener's left auditory cortex followed the hierarchical organization for speech processing, with response lags in A1+ (~0 s), STG/STS (~2 s), and MTG (~4 s) increasing linearly. Such auditory–articulatory neural coupling substantially diminished when speech comprehension failed. Moreover, stronger coupling

responses in the STG/STS were associated with better speech comprehension. Implications of these findings are discussed below.

Reference to the Articulatory Motor Information Facilitates Speech Comprehension

Whether speech perception and comprehension rely on references to the articulatory motor information of the speaker is a long-standing debate. A range of studies have revealed the involvement of motor cortex during speech perception tasks (e.g., Pulvermüller et al. 2006; Möttönen and Watkins 2009; Du et al. 2016). However, given the nature of multiple functions of human brain (Bassett and Gazzaniga 2011), it is difficult to determine whether the response of motor cortex in a listener's brain is associated with speech articulation or with other nonspeech related functions such as task-specific motor control (Hickok 2009, 2014). Instead, the current study directly examined the linkage of listener's auditory system with speaker's articulatory system during verbal communication. It is the first time, to our knowledge, significant couplings between the listener's auditory system and speaker's articulatory system were revealed. Furthermore, listener's coupling responses were positively correlated with their speech comprehension. These findings provide novel and reliable evidence supporting that auditory speech processing is facilitated by references to the articulatory motor information of the speaker. Yet, whether and

how such between-brain auditory-articulatory coupling might contribute to “sound to articulation” transformation within listener’s brain need to be further investigated.

Temporal Profile of Coupling Response in Listener’s Auditory Cortex

We observed that the lags of coupling responses in listener’s left A1+ (~0 s), STG/STS (~2 s), and MTG (~4 s) are increasing linearly. Such a spatial-temporal profile indicates an information flow from lower-order, acoustic-processing areas to higher-order and semantic-processing areas during speech comprehension. This spatial-temporal profile is in line with a previous study that examined the activity within listener’s brain during narrative comprehension using MEG (Ding and Simon 2012), yet the time scales of the current results are much longer, especially for the responses in the left MTG. The relatively long lag might be partly due to the tardiness of blood oxygen level-dependent (BOLD) signals and the low sampling rate of fMRI. Another cause might be that the inter-brain neural coupling reflects slow processes (such as the construction of sentence-level meaning or the situation model) that are characterized by larger time scale. Indeed, other studies investigating neural coupling between communicating pairs have also reported comparably long lags in the listener’s brain (Stephens et al. 2010; Silbert et al. 2014). In particular, a study using time-sensitive electroencephalography (EEG) observed that the components contributing predominantly to the inter-brain similarities of listeners are those with low frequency (~0–1 Hz). Further, the coupling responses in the listener’s brain to the speaker’s brains peaked at a lag of 12 s (Kuhlen et al. 2012).

To have an intuitive sense of the response lag, we conducted a detailed analysis (see [Supplementary Material](#)) on the stories. We found that 4 s correspond approximately to a time span in which the speaker produced a “narrative unit” that expresses one or more propositions (e.g., “我给大家讲一个我第一次坐军舰的事情\ I’d like to share you a story about my first time to take a cruise”). Presumably, the lag of 4 s in listener’s left MTG results from the time difference between the large narrative unit being planned and produced by the speaker, relative to when it becomes comprehended by the listener.

Potential Driving Sources for the Auditory-Articulatory Neural Coupling between Listener and Speaker

A set of results suggest the auditory-articulatory neural coupling we observed is not driven merely by shared sensory-level inputs (such as the presence of sound) between the listener and speaker but depends on the transfer of multiple-level linguistic information. First, an orderly temporo-spatial progression was observed in listener’s left auditory cortex when listening to the intelligible Chinese stories: the A1+ of listener showed coupling response immediately after the vocalization of the speaker; then 2 s after the vocalization, the STG/STS showed coupling response, and 4 s after the vocalization, the MTG showed coupling response. Such coupling responses follow the hierarchical organization for the processing of meaningful speech (Pelle et al. 2010). In contrast, when listening to the unintelligible Mongolian speaker, despite of shared sensory-input with the speaker, the listener’s coupling responses vanished substantially. Further, listeners showing stronger coupling response to the speaker understood the story better. Taken together, merely

shared sensory inputs between the listener and speaker would be difficult to account for the delayed responses observed in STS/MTG, nor the relevance to speech comprehension.

Among the linguistic information of natural speech, we assume the quasi-rhythmic information at multiple temporal scales (such as at the phoneme, syllable, word, or semantic unit levels) is one of the critical elements that enables the between-brain auditory-articulatory neural coupling to occur. Indeed, it is suggested that, during the evolution of human speech, the articulatory motor system has structured its output to match those rhythms the auditory system can best apprehend; similarly, the auditory system has likely become tuned to the complex acoustic signal produced by the articulator’s rhythmic movements (Giraud and Poeppel 2012). The auditory-articulatory neural coupling we observed may partly build upon both alignments of listener’s auditory system and of speaker’s articulator system to the quasi-rhythmic temporal features of speech. In addition to the quasi-rhythmic information, the multi-timescale prosodic information of natural speech (such as intonation, tone, and stress) is likely to be another critical enabling element. Speech prosody contains rich information that is important for one to accurately construct the meaning of large speech units. For instance, using a different tone to utter the last word of a sentence could convey opposite meanings. Previous studies have shown that the MTG are involved in the construction of sentence-level meaning (Humphries et al. 2007; Mashal et al. 2009). Given that the generation of prosody is tightly associated with the activity of larynx (Isshiki 1964), we assume the need to utilize prosodic information for the construction of narrative meaning may lead to the coupling responses of listener’s MTG to speaker’s LPA.

Selective Coupling Responses to the LPA

Among the articulatory ROIs examined, only bilateral LPA in the speaker was consistently coupled to the auditory cortex in listeners. The consistent coupling of the LPA with the auditory cortex could be explained by its potential function in controlling the movement of larynx. The larynx is the articulatory organ for vocalization, determining the volume and pitch of the produced sound (Isshiki 1964); therefore, neural dynamics of speaker’s LPA may be tightly linked to the temporal fluctuation and prosodic features of the produced speech and coupled with the neural dynamics in listener’s auditory cortex, which encodes such information of the input speech (Ding and Simon 2012; Ding et al. 2016). In contrast, the movement of the tongue modifies rather than generates vocal sounds, influencing the fine-grained spectrotemporal structure of the produced speech. We assume activity of speaker’s “tongue” areas may be related to the activity of listener’s auditory cortex at a more microscopic level beyond the current fMRI resolution. The left frontal operculum, which is involved in planning rather than execution of articulation (Price 2010), may not directly determine the auditory properties of the produced speech and, hence, is unlikely to be significantly coupled with the listener’s auditory cortex.

Limitations

The current study has some limitations. First, when calculating listener’s coupling responses at different lags, the auto-correlations of BOLD signal were not controlled. This would partly lead to the overlap of coupling maps across lag 0 s, lag 2 s, and lag 4 s. Though it is the change rather than the overlap of

coupling maps across lags that interests us, and a quantification for the auto-correlation shows this effect is small (Supplementary Fig. S8), future studies should have the auto-correlation controlled to obtain lag-specific coupling maps. Secondly, the current study used Mongolian story as the condition controlling for the intelligibility of speech. However, in addition to intelligibility, other unmatched aspects (such as attention and the tonal and nontonal nature of Chinese and Mongolian) may also contribute to the differences in listener's coupling responses to the Mongolian speaker and to the Chinese speaker. More strict experimental control is needed to better understand the exact driving sources for the inter-brain neural coupling.

Conclusion

In summary, this study demonstrates that neural dynamics in listener's auditory cortex are inherently aligned with the dynamics in speaker's articulatory cortex during verbal communication. This auditory-articulatory neural alignment is associated with linguistic information transfer from the speaker to the listener and facilitates listener's comprehension for the speech. These findings provide reliable evidence supporting that auditory processing of speech relies on references to the articulatory motor information of the speaker and open a new window into the neural basis underlying verbal communication.

Supplementary Material

Supplementary material is available at *Cerebral Cortex* online.

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